

AMyanHlu — Functional Specification

Donor, Staff & Administrator Functions

Project: AMyanHlu

Document Purpose: Establish a common functional understanding among all development team members.

1. SYSTEM ROLES

AMyanHlu has three primary authenticated roles:

1. DONOR
 2. STAFF
 3. ADMIN

1.1 Donor

A donor is a registered blood donor who uses the mobile application to manage their blood donation activities.

- The donor CAN:**

 - Manage their account and profile
 - Manage authentication
 - View their donor information
 - Use their NFC donor card
 - Search and select hospitals
 - Complete hospital-specific questionnaires
 - Book donation appointments
 - Manage appointments
 - View donation history
 - View blood test results
 - View certificates
 - Read news/articles
 - Request permission to organize donation events
 - Use the AI assistant
 - Receive notifications

- The donor CANNOT:**

 - Modify verified blood type directly
 - Verify their own donation
 - Check themselves in/out as an official hospital record
 - Issue certificates
 - Issue blood test results
 - Create hospital questionnaires
 - Approve donation-event requests
 - Search emergency donors
 - Manage other users

2. DONOR FUNCTIONS

D01 — Donor Registration

Purpose: Allow a new donor to create an AMyanHlu account.

Actor: DONOR

Preconditions: Donor does not already have an AMyanHlu account; Donor has a valid phone number.

Process:

Open Registration → Enter phone number → Verify phone number → Enter personal information → Enter NRC information → Upload NRC front/back → Set password → Accept terms → Submit registration → Create donor account

Information collected: Name, Phone number, Email (optional/required depending on final decision), Date of birth, Gender, NRC, Address, Blood type (if known).

Result: A donor account is created with initial account status ACTIVE. Medical information that has not yet been verified must be treated as **unverified**.

D02 — Donor Login

Purpose: Allow donors to access the application securely.

Authentication methods: Password, Biometric authentication, or both.

Password Login: Phone/Email + Password → Backend authentication → Access granted
Biometric Login: Performed by mobile OS. Server must not store raw biometric data.

Result: The donor enters the authenticated application.

D03 — Donor Profile Management

Purpose: Allow donors to view and update their personal information.

Donor can update: Name (where permitted), Phone/email (according to verification rules), Address, Profile picture, Other non-medical profile information.

Restricted information: Verified blood type, Verified NRC information, Donation history, Medical test results, Certificates (controlled by hospital staff/system processes).

D04 — Change Password

Purpose: Allow the donor to change their account password.

Current password → New password → Confirm new password → Validate → Hash new password → Update account

Rule: Passwords must never be stored as plaintext.

D05 — Enable/Disable Biometric Authentication

Purpose: Allow the donor to use device biometric authentication for application access.

Donor enables biometric login → Device verifies biometric → Application enables biometric authentication

D06 — View Donor Card

Purpose: Allow the donor to view information about their NFC donor card.

Information displayed: Card status, Card issue date, Card expiry date, Issuing hospital, Card identifier (if appropriate).

Possible statuses: ACTIVE, EXPIRED, LOST, BLOCKED, REPLACED.

D07 — Report Lost NFC Card

Purpose: Allow a donor to report that their NFC card has been lost.

Donor selects "Report Lost Card" → Confirmation → Card status → LOST

D08 — View Hospitals

Purpose: Allow donors to find hospitals where they can donate.

Functions: View hospital list, Search hospital, View hospital details (Name, Profile image, Address, Phone, Email, Operating hours, Donation service info), Select hospital for appointment.

D09 — Select Preferred Hospital

Purpose: Allow the donor to choose a hospital where they want to donate. Establishes the hospital associated with the donor's appointment/questionnaire process without permanently transferring the donor.

D10 — View Hospital Questionnaire

Purpose: Allow the donor to complete the self-screening questionnaire required by the selected hospital.

Select hospital → System finds published questionnaire → Display questionnaire → Donor answers questions

Rule: Only questionnaires with status = PUBLISHED should be presented.

D11 — Complete Self-Questionnaire

Purpose: Collect pre-donation information from the donor.

Question types: YES / NO, Single choice, Multiple choice, Text, Number, Date.

Open questionnaire → Answer questions → Validate required fields → Submit → Store response

Important Rule: Pre-donation screening only; not a medical diagnosis. Final medical decision remains with hospital staff.

D12 — Save Questionnaire Draft

Allow donors to leave an incomplete questionnaire (DRAFT) and continue later. Becomes SUBMITTED only upon explicit submission.

D13 — Book Donation Appointment

Purpose: Allow a donor to reserve a donation appointment at a selected hospital.

Select hospital → Select date → View available time slots → Select time → Complete questionnaire → Review information → Confirm appointment

Appointment Statuses: PENDING, CONFIRMED, CANCELLED, COMPLETED, MISSED, REJECTED.

D14 — View Upcoming Appointment

Display hospital, date, time, appointment status, questionnaire status, and relevant instructions.

D15 — Cancel Appointment

Open appointment → Select Cancel → Confirm cancellation → Enter reason (if required) → Appointment → CANCELLED

D16 — Reschedule Appointment

Existing appointment → Select Reschedule → Select new date/time → Confirm → Update appointment

D17 — View Donation History

Display donation date, hospital, donation type, donation status, related medical records, certificate availability, and next eligible donation date.

D18 — View Donation Eligibility

Calculates preliminary eligibility based on previous donations, intervals, card status, deferrals, and questionnaire results. Statuses: ELIGIBLE, NOT_ELIGIBLE, REQUIRES_REVIEW.

D19 — View Next Eligible Donation Date

Last completed donation + 4 months → Next eligible date

D20 — View Blood Type

View verified blood type (e.g. B+) or unverified status (e.g. B - Status: Unverified). Donor cannot directly modify verified blood type.

D21 — View Blood Test Results

Access blood-test results issued by authorized hospital staff (Test date, Hospital, Blood type, Laboratory results, Document).

D22 — View Certificates

View/download certificates (Donation Certificate, Participation Certificate) issued by hospitals.

D23 — Read News and Articles

Browse and read articles published by Hospital Staff or Administrators. Donors cannot create or edit articles.

D24 — Request Blood Donation Event

Allow donors to request permission to organize an event. Statuses: PENDING, APPROVED, REJECTED, CANCELLED.

D25 — View Event Request Status

Track request status and review rejection reasons provided by hospital staff.

D26 — AI Assistant

Provide conversational assistance regarding blood donation, preparation, FAQs, and app usage. Must NOT act as a replacement for a doctor or make medical diagnoses.

D27 — Notifications

Receive alerts for appointments, test results, certificates, NFC card status, eligibility updates, event requests, and announcements.

3. STAFF FUNCTIONS

Business Rule: Staff accounts are associated with exactly one hospital (One Hospital = One Staff Account). Operations are strictly restricted to their assigned hospital.

S01 — Staff Login

Authenticate using password and/or biometric authentication based on configuration.

S02 — View Hospital Dashboard

Operational overview: Today's appointments, expected donors, checked-in donors, completed donations, pending questionnaires, pending event requests, emergency blood requests.

S03 — Search Donor

Search donors by Donor ID, Name, Phone number, NFC card UID, NRC, or Blood type for legitimate hospital operations.

S04 — NFC Donor Check-In

Donor arrives → NFC card scanned → System identifies donor → Verify card → Find appointment → Calculate/display eligibility → Staff reviews result → Staff confirms check-in

Note: NFC check-in does NOT mean the donor has successfully donated blood.

S05 — Handle Donor Screening

Review pre-screening questionnaire responses, donation history, and medical records before donation.

S06 — Record Donation

Record actual donation procedure. Statuses: STARTED, COMPLETED, FAILED, CANCELLED, DEFERRED. Only COMPLETED updates donation history/eligibility.

S07 — NFC Donor Check-Out

Donor finishes process → NFC card scanned → Find active visit → Staff confirms result → Check-out recorded

S08 — Create/Upload Questionnaire

Method A (Manual): Add questions, configure types, set required, publish.

Method B (AI/OCR): Upload questionnaire image → OCR/AI processing → Draft created → Staff edits & publishes.

S09 — Edit Questionnaire

Modify questions, types, options, order. Published questionnaires should use versioning.

S10 — Publish Questionnaire

DRAFT → PUBLISHED → ARCHIVED

S11 — Review Questionnaire Responses

View submitted answers, flag problematic responses, and mark as reviewed.

S12 — Verify/Update Blood Type

Verify donor's blood type using authorized test results. Updates current blood type, creates history record, and writes audit log.

S13 — Issue Blood Test Result

Upload/issue official test results visible to the donor.

S14 — Issue Certificate

Generate/upload certificates with unique certificate numbers for completed donations or events.

S15 — Review Donation Event Requests

Review donor event applications and decision as APPROVE or REJECT with review notes.

S16 — Manage Donation Events

Manage approved events. Event statuses: UPCOMING, ONGOING, COMPLETED, CANCELLED.

S17 — Create Emergency Blood Request

Create request with blood type, units required, urgency (NORMAL, URGENT, CRITICAL), reason, and expiration.

S18 — Search Eligible Donors for Emergency Need

Staff-only search filtering eligible donors based on blood type matching, active status, donation interval, and location availability.

S19 — Donor Location Availability

Utilize opt-in location sharing (ON / OFF) for emergency donor matching. No continuous tracking by default.

S20 — Staff Notifications

Receive notifications for appointments, arrivals, event requests, emergency requests, and questionnaires.

4. ADMIN FUNCTIONS

Administrators have system-wide privileges and are not restricted to a single hospital.

A01 — Admin Login

Secure login with high security enforcement.

A02 — Admin Dashboard

System-wide metrics: Total donors, active staff, hospitals, appointments, completed donations, emergency requests, system activity.

A03 — Manage Donors

Search, activate, deactivate, suspend donor accounts, and assist with registration.

A04 — Create Donor Account

Register donors unable to use the app independently. Temporary credentials provided securely.

A05 — Manage Staff

View, activate, suspend, or deactivate hospital staff accounts.

A06 — Create Staff Account

Register new hospital and create its single staff account. Database enforces 1-to-1 relationship.

A07 — Manage Hospitals

Create, edit, activate, deactivate, or suspend hospital records.

A08 — Manage Articles

Create, edit, publish, or archive platform educational articles and news.

A09 — Manage System-Wide News

Publish platform announcements, campaign info, and app update notices.

A10 — Manage Blood Type Reference Data

Maintain controlled reference list (A+, A-, B+, B-, AB+, AB-, O+, O-).

A11 — Monitor System Activity

View system activity logs for account creation, blood type changes, card operations, certificates, etc.

A12 — Audit Logs

Immutable logging of staff/admin actions (e.g., Staff ID, Action, Target Donor, Old Value, New Value, Timestamp).

A13 — Manage System Notifications

Broadcast system-wide notices for maintenance, policy updates, and alerts.

5. CROSS-ROLE FUNCTIONAL RULES

- **R01 — One Account Per Donor:** Each donor has exactly one account (One Donor = One Account).
- **R02 — One Staff Account Per Hospital:** Enforced by database unique constraint (One Hospital = One Staff Account).
- **R03 — Restricted Blood Type Changes:** Donors cannot edit blood types. Only authorized staff can verify/update.
- **R04 — Donation ≠ Check-In:** Arrival/Check-in does not imply successful donation.
- **R05 — Donation ≠ Check-Out:** Check-out does not complete donation. Explicit COMPLETED status required.
- **R06 — Donation History Source of Truth:** Derived from donations table records, not a standalone text field.
- **R07 — System-Calculated Eligibility:** Calculated automatically; final authority rests with hospital medical staff.
- **R08 — NFC for Identification:** Cards identify donors; backend maintains state and medical records.
- **R09 — Hospital-Specific Questionnaires:** Questionnaires are tied strictly to the destination hospital.
- **R10 — Human Review for AI Drafts:** AI/OCR questionnaires require staff review before publishing.
- **R11 — Hospital-Scoped Staff Access:** Staff accounts are restricted to their assigned hospital's data.
- **R12 — Admin Scope:** System-wide access subject to audit trails.

6. COMPLETE ROLE COMPARISON

Function	Donor	Staff	Admin
Register donor	Self	No	Yes
Login	Yes	Yes	Yes
Manage own profile	Yes	Yes	Yes
Manage donors	No	Limited	Yes
Manage staff	No	No	Yes
Manage hospitals	No	No	Yes
View own NFC card	Yes	Yes	Yes
Issue NFC card	No	Hospital process	Manage / authorize
NFC check-in	No	Yes	Monitoring
NFC check-out	No	Yes	Monitoring
Book appointment	Yes	On behalf if needed	On behalf if needed
Cancel appointment	Yes	Yes	Yes
Create questionnaire	No	Yes	System-level if needed
Edit questionnaire	No	Yes	Yes
Publish questionnaire	No	Yes	Yes
Answer questionnaire	Yes	No	No
Review questionnaire	No	Yes	Authorized oversight

Function	Donor	Staff	Admin
Record donation	No	Yes	Authorized oversight
View donation history	Own	Hospital donors	System-wide
Verify blood type	No	Yes	Authorized oversight
Change verified blood type	No	Yes	Authorized oversight
Issue blood test	No	Yes	Authorized oversight
View blood test	Own	Hospital donors	Authorized
Issue certificate	No	Yes	Authorized
View certificate	Own	Yes	Authorized
Create article	No	Yes	Yes
Publish system news	No	Hospital content	Yes
Request donation event	Yes	No	No
Review event request	No	Yes	Yes
Create emergency request	No	Yes	Yes
Search emergency donors	No	Yes	Authorized admin
Location matching	Opt-in	Yes	Authorized
AI assistant	Yes	Optional	Optional
Manage notifications	Own	Hospital	System-wide
Audit logs	No	Limited	Yes

7. END-TO-END DONOR LIFECYCLE

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REGISTER → ACCOUNT CREATED → NRC / IDENTITY INFO → NFC CARD ISSUED → CARD LINKED TO DONOR
↓
DONOR SELECTS HOSPITAL → COMPLETES QUESTIONNAIRE → BOOKS APPOINTMENT → APPOINTMENT CONFIRMED
↓
DONOR ARRIVES → NFC CHECK-IN → ELIGIBILITY DISPLAYED → HOSPITAL SCREENING → DONATION
↓
DONATION COMPLETED → NFC CHECK-OUT → RECORD STORED → BLOOD TEST / CERTIFICATE ISSUED
↓
DONOR VIEWS RECORDS → NEXT ELIGIBLE DATE CALCULATED (+4 months) → ELIGIBLE TO DONATE AGAIN

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8. EMERGENCY DONOR MATCHING LIFECYCLE

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HOSPITAL creates emergency request (Type + Units + Urgency)
↓
SYSTEM searches eligible donors (Blood type match + Eligibility + Active + Location opt-in)
↓
STAFF receives matching list → Contacts donors according to hospital protocol

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9. DONATION EVENT LIFECYCLE

DONOR creates request → PENDING → HOSPITAL STAFF reviews → APPROVED / REJECTED
↓ (If Approved)
EVENT CREATED → UPCOMING → ONGOING → COMPLETED

10. QUESTIONNAIRE LIFECYCLE

STAFF creates manually OR uploads image (AI/OCR) → Draft questionnaire → Staff review/edit
↓
DRAFT → PUBLISHED → Donor answers → Response submitted → Staff reviews → REVIEWED → ARCHIVED

11. BLOOD TYPE VERIFICATION LIFECYCLE

DONOR current type: B (Unverified) → Hospital blood test → Result: B+
↓
STAFF VERIFICATION → Current type updated to B+ → Create history record → Write audit log →
Donor sees: B+ VERIFIED

12. IMPORTANT TERMINOLOGY

Donor	A registered person who can donate blood.
Account	Authentication identity used to access the system.
Staff	Authorized hospital-side user responsible for operational functions.
Hospital	Medical institution associated with staff, appointments, questionnaires, and donations.
NFC Card	Physical donor identification card issued by a hospital.
Appointment	Scheduled intention to visit a hospital for donation.
Check-In	Recorded arrival and verification of a donor at a hospital.
Screening	Hospital assessment of whether the donor can proceed with donation.
Donation	Actual blood donation procedure/record.
Check-Out	Recorded completion of the donor's hospital visit.
Completed Donation	Donation explicitly recorded as successfully completed.
Eligibility	Preliminary system determination based on configured rules.
Questionnaire	Hospital-defined set of pre-donation screening questions.
Donation Event	Planned blood donation activity involving multiple participants.
Blood Request	Hospital requirement for blood of a specific type and quantity.

13. FEATURES EXPLICITLY NOT INCLUDED IN DONOR APP

Out of Scope for Donor App (To prevent scope confusion):

- Emergency blood search
- Emergency donor matching
- Staff management
- Hospital management
- NFC verification
- Blood type verification

- Blood test issuing
- Certificate issuing
- Questionnaire creation/publishing
- Donor management
- System-wide administration
- Audit log management

14–16. CORE APPLICATION NAVIGATION

Customer App Nav

HOME | Eligibility
| Upcoming Appt |
Next Date | Quick
Actions HOSPITALS |
Find/Select
APPOINTMENTS | Book /
Cancel | Reschedule
MY DONATION | History
| Test Results |
Certificates MY NFC
CARD | Card Status AI
ASSISTANT NOTIFICATIONS
PROFILE / SETTINGS

Staff App Nav

DASHBOARD DONORS |
Search / View
APPOINTMENTS |
Today's List CHECK-IN /
OUT | NFC Scanner
DONATIONS | Records
QUESTIONNAIRES |
Create / Edit | AI
Generate | Responses
MEDICAL RECORDS |
Tests / Certs EMERGENCY
BLOOD ARTICLES

Admin App Nav

DASHBOARD DONORS |
Search / Create STAFF |
Create / Manage HOSPITALS
| Create / Manage
ARTICLES | Create /
Publish EMERGENCY BLOOD
SYSTEM NOTIFS AUDIT LOGS
SYSTEM SETTINGS

17. OVERALL RESPONSIBILITY MODEL

DONOR ("I want to donate.")
Select Hospital → Answer Questions → Book Appointment → Visit Hospital → Show NFC Card

STAFF ("This donor has arrived.")
Scan NFC → Verify donor → Check eligibility → Perform screening → Record donation → Issue medical records/ certificate

ADMIN ("I manage the platform.")
Manage hospitals → Manage staff → Assist with donor accounts → Manage system content → Monitor system activity

14. Core Customer App Navigation

HOME

- └ Donor Eligibility
 - └ Upcoming Appointment
 - └ Next Eligible Date └
- Quick Actions

HOSPITALS

- └ Find / Select Hospital

APPOINTMENTS

- └ Upcoming
- └ History
- └ Book
- └ Cancel
- └ Reschedule

MY DONATION

- └ Donation History
 - └ Blood Test Results └
- Certificates

MY NFC CARD

- └ Card Status

QUESTIONNAIRES

- └ Self-Screening

NEWS

- └ Articles

DONATION EVENTS

- └ Request Event

AI ASSISTANT

- └ Blood / Donation / App Help

NOTIFICATIONS

PROFILE

SETTINGS

15. Core Staff App Navigation

DASHBOARD
DONORS Search / View
APPOINTMENTS Today's Appointments
CHECK-IN / CHECK-OUT NFC Scanner
DONATIONS Donation Records
QUESTIONNAIRES - Create - AI Generate - Edit - Publish - Responses
MEDICAL RECORDS - Blood Tests - Certificates
EVENT REQUESTS
EMERGENCY BLOOD
ARTICLES
NOTIFICATIONS
PROFILE / SETTINGS

16. Core Admin Navigation

DASHBOARD
DONORS - Search - Create - Manage - Status
STAFF - Create - Manage - Status
HOSPITALS - Create - Manage - Status
ARTICLES - Create - Publish - Manage
EMERGENCY BLOOD
SYSTEM NOTIFICATIONS
AUDIT LOGS
SYSTEM SETTINGS
PROFILE / SETTINGS